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Research Program/Project/Study Title: Development of Computer Simulated Biomass-Fired Multilayered Fish Dryer	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	
		Province:	
		Region:	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity		(8) Discipline/Field of Specialization	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

This study was conducted to develop a multilayered biomass-fired fish dryer. The initial design was subjected to computer simulation prior to fabrication to ensure uniformity of drying air in the chamber in terms of air flow rate and temperature. The biomass-fired fish dryer was composed of biomass furnace, heat exchanger, drying chamber, and blower.

The performance of the fabricated dryer was tested in terms drying air temperature, drying efficiency, combustion efficiency and heating system efficiency. Drying air temperature ranged from 69°C to 87°C. At the target moisture content of the dried fish set to 25%, the mean drying time was 496.25 minutes or 8.27 hours. The drying efficiency was 66.97% considering the heat load required to vaporize moisture from fish and the heat utilized in drying. The heating system efficiency was 55% considering the heat supplied by the burner, the fuel consumption, and the heat content of the fuels.

A total energy cost of Php26.00 per batch of 40 kg of fish was spent. The computer simulation using Computational Fluid Dynamics aided in enhancing the design of the drying chamber of the biomass-fired dryer, thereby improving the air flow uniformity. The dryer was more efficient than the existing biomass-fired dryer of CAPSU. Studies to increase the capacity of the dryer from 40 kilograms to 100 kilograms per batch is also forwarded.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

fish dryer, simulation, drying, biomass furnace

Fish is an important foodstuff for its nutritional value as a main source of protein. In the Philippines, the mean per capita consumption of fish and fishery products in 2017 was 40 kg/year or 109 grams/day, representing 12.8% of the total food intake. Based on the SEAFDEC report by Lamarca N. (2017), around 30% of the country's fish catch is being processed since it is a highly perishable food that invariably becomes rotten within a few hours after capture. "In the Philippines, fish drying is the most practiced processing method, especially for the small-sized species", (Lamarca N., 2017). It was found that reducing the moisture content of fish to 25% will retard bacteria and reduce autolytic activity.

Sun-drying is the widely used method for fish drying as it requires cheaper operating expenses. However, this method has disadvantages due to unpredictable weather conditions, resulting in uneven drying and a relatively slow drying rate, thus the tendency of spoilage. Hence, several dryers have been developed to dry fish products. Dryers that use a combination of solar and biomass as a source of energy have emerged. Also, the "use of LPG/electricity has substantially increased in the last few years" for the said purpose (Patchimaporn, U., *et al*, 2020). However, these drying heat sources are associated with higher initial and operating costs. Thus, they have not been considerably used in the industry.

Biomass is an effective alternative to traditional sun-drying and dryers, which use expensive electricity and fossil fuels for their heat source. It has been used successfully as a fuel in many agricultural machines – including a flatbed dryer for grains. Rice husk, a by-product of rice milling, for example is an attractive biomass fuel because they are readily available in large quantities and easy to collect.

The prior-arts on fish drying systems involve different designs and encompass a common principle: the "hot air absorbs and conveys moisture from the product by heat and mass transfer and diffusion while the residual air is discharged through an exhaust unit." The most common types are cabinet dryers consisting of several layers of fish trays. Non-uniformity of dried products in terms of final moisture content is a common problem with this type of dryer. Flatbed dryer is another type and was introduced for higher drying efficiencies. Ideally, however flatbed dryers are designed to dry materials with smaller surface areas, porous, and have initial moisture contents (MC) of 25 to 30%, such as grains. In the case of fish with higher MC (up to 80%) and larger surface areas, case hardening is a disadvantage. Due to higher space requirements, the capacity (load/batch) would be another limitation in this type of dryer. Therefore, this study was conducted to modify the designs of the existing dryers and use computer simulation as a tool to enhance the design prior to fabrication and actual testing.

The technology developed from this study may benefit the dried fish processors-cum-traders in the province and its neighboring provinces by providing a drying machine that is hygienic and efficient. Essentially, it will also be helpful to fisherfolks by ensuring fish catch demand even during the rainy season and to the consumers as it may significantly stabilize dried fish prices.

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Generally, this study was conducted to develop a biomass-fired fish dryer. Specifically, it aims to:

1. design a multi-layer biomass-fired fish dryer and modify the design using simulation;
2. fabricate a prototype dryer;
3. determine the performance of the dryer; and
4. determine cost and return of fish drying in using the developed dryer.

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Design Considerations

The area of the drying trays and the chamber was designed considering the surface area of fish, particularly *Manamsi (Sardinella Fortunella)* fish. On the other hand, the design of the heating system and fan was based on the amount of moisture to be removed from the 40-kilogram fish input, the air flow requirement, calculated using the formula:

$$W_w = W_{i1} - \frac{1-MCi}{1-MCf}$$
$$A_f = \frac{Ww}{AHf - AHi}$$

The dryer was designed to dry the fish input within 7-8 hours. Thus, the heat requirements were computed using the following formula:

Heat = Af X Enthalpy Difference

Energy_r= heat / drying time

Burner Capacity = Heat Req. / (Energy Content of fuel X Eff X 3600 sec/hr)

Fuel Consumption = Burner Capacity X Total Drying Time

Simulation using Computational Fluid Dynamics (CFD) was performed with the initial dryer design. Air velocity, temperature, and pressure drop along the dryer were assessed. The simulation was performed using SOLIDWORKS Flow Simulation application. To enhance drying air characteristics in the drying chamber, design iterations were made.

Technical Description of the Multi-Commodity Biomass-Fired Dryer

The biomass-fired fish dryer (Appendix 1) was composed of following components: biomass furnace, heat exchanger, drying chamber, and blower.

Biomass furnace. An indirect-type biomass furnace was constructed as the dryer’s heat source. It was composed of a biomass bin with a feeding hopper a combustion chamber, an ash collector, a blower and a chimney. The rice hulls were fed into the feeding hopper to the combustion chamber for burning. The blower is installed to supply air and ensure continuous combustion. The chimney carries undesirable combustion products.

Heat exchanger. A heat exchanger was constructed in the biomass furnace to prevent product of combustion to come directly to the product being dried. It was composed of several metal pipes. The first open-end of the GI pipes is exposed to the ambient air for air suction while the other-end were connected to a duct.

Drying chamber. The biomass-fired fish dryer was composed of a number of drying layers, plenum chambers, drying trays and an exhaust chimney. The drying chamber were insulated. The drying trays were made of stainless mesh wires. The parts were held by a support stand.

Blower. A blower was used to force heated air through the drying chamber. It was connected to a duct on one end and with other end connected to the heat exchanger to carry heated air from the heat exchanger to the drying chamber.

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(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Simulation

Results of computer simulation of the original design/set-up revealed that there was an uneven distribution of air in the dryer in terms of air flow rate and temperature (Appendix 2). The maximum temperature was 98.87°C and the minimum was 89.78 °C (Appendix 3). The rightmost lower region of the drying chamber received a very low air flow velocity of only 0.054 m/s, while the rightmost upper portion received the highest air flow velocity of 3.424 m/s (Appendix 4). It was found that this partly resulted from the formation of pressure inside the drying chamber. To enhance the air flow uniformity in the drying chamber, the following were modified in the design:

1. Addition of exhaust outlet on the end – most part of the dryer to avoid vacuum.
2. Provision of baffle on the connecting duct to reduce the pressure build – up.
3. Increasing the air duct length to 1.5 meters to lessen pressure build – up on the transition duct.

Appendix 5 shows the simulation results of the final drying design. The addition of the baffles, the lengthened air duct and the modified exhaust design improved the air flow distribution inside the drying chamber. Thus, the final design was used in the fabrication of the machine.

Drying Air Temperatureat 30-Minute Interval

Appendix 6 shows the mean drying air temperature measured at eight different drying trays at 30-minute interval. Drying air temperature ranged from 69°C to 79°C. It can be observed that there were minimal variations in the drying air temperatures in the drying trays and during the time intervals. The graph also shows that the drying air temperature was not influenced by the time of the day. The variations in the temperature can be attributed to the gravity-controlled flow of rice hull fuels to the combustion chamber thereby producing unequal rate of combustion and energy supplied over time.

Drying Time and Capacity

Appendix 7 presents the data on fish drying time and dying rate of the developed dryer. At the target moisture content of the dried fish set to 25%, the mean drying time was 496.25 minutes or 8.27 hours. Results of the test also revealed that drying capacity was 4.87 grams per hour. The mean moisture loss was 27 kilograms per batch of 40-kilograms.

Drying Performance of the Biomass Fish Dryer

The drying performance of the biomass-fired fish dryer in terms of drying efficiency, combustion efficiency, and heating system efficiency are reflected in Appendix 8. The drying efficiency was 66.97% considering the heat load required to vaporize moisture from fish and the heat utilized in drying. The heating system efficiency was 55% considering the heat supplied by the burner, the fuel consumption, and the heat content of the fuels. This was higher than the required heating system efficiency for indirect-fired dryer which is 50% based on Philippine Agricultural Engineering Standards. This was also higher than CAPSU’s 20-kilogram fish dryer with only 49% efficiency. It can be attributed to the higher airflow rate through some modifications in the drying chamber, blower, exhaust, air duct, and combustion chamber.

Fuel and Energy Consumption of the Biomass-Fired Fish Dryer

The mean weight of rice hulls consumed was 73.5 kilograms for the whole drying process. This means that at PHP10/bag of 10kg of rice hulls, the cost of fuel for the whole drying process of 40-kilogram fish was Php73.50. The electrical energy consumption for the blower was 2.00 kW-hr/batch. This means that at Php13.00 energy cost per kW-hr, a total energy cost of Php26.00 per batch of 40 kg of fish was spent.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

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CONCLUSIONS Simulation using Computational Fluid Dynamics can be used to enhance the design of the drying chamber of the biomass-fired dryer, thereby improving the air flow uniformity. The biomass-fired fish dryer is suitable for drying of <i>Manamsi</i> fish within the required drying temperature for fish of 70°C to 80°C with final moisture content of 25%. It can used even at night or during bad weather conditions. The dryer was more efficient than the existing biomass-fired dryer of CAPSU. RECOMMENDATIONS Considering the advantages in terms of drying performance of the biomass-fired fish dryer, it is recommended that the technology be introduced for pilot-testing by the dried-fish processors and be tested to other kinds of fish. Studies to increase the capacity of the dryer from 40 kilograms to 100 kilograms per batch is also forwarded.
(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)
(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)
(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.
(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)
(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.) Aderibigbe, D. A.V, Adesola K., & Obayopo., S. O.,(2016). <i>Journal of Engineering and Technology</i>, Volume 1, Issue 1, September 2016 http://engineering.fuoye.edu.ng/journal/index.php/engineer/article/download/22/pdf Adeoye, O. A., <i>Department of Agricultural and Biosystems Engineering, Faculty of Engineering and Technology, University of Ilorin, P. M. B.1515, Ilorin 240003, Kwara State, Nigeria</i> Apaga, A.R.M., Hipolito, A.V., & Martinez, H.F. (2008). <i>Techno-Guide on Multi-Commodity Solar Tunnel Dryer (MCSTD)</i>http://www.mixph.com/techno-guide-on-multi-commodity-solar-tunnel-dryer-mcstd Department of science and technology (DOST). (2015). Solar fish dryer to help small fishermen http://www.dost.gov.ph/knowledge-resources/news/44-2015-news/858-dost-s-solar-powered-fish-dryer-to-help-small-fishermen.html Gunasekaran, K., Shanmugam, V., & Suresh, P. (2012). Modeling and Analytical Experimental Study of Hybrid Solar Dryer Integrated with Biomass Dryer for Drying Coleus Forskohlii <i>Stems IPCSIT</i> vol. 28 2012 and 2012 IACSIT Press,Singaporehttps://pdfs.semanticscholar.org/1e95/885457cdc2b842c4aac26913d59d1ded1c1b.pdf Gunasekaran, K., & Suresh, P. (2012). Department of Mechanical Engineering, Muthayammal Engineering College, Rasipuram, Namakkal Dist – 637 408 Lamarca N. (2017) Fisheries Country Profile: Philippines, 2017 Regional Fisheries Policy Network (RFPN) Member for the Philippines, http://www.seafdec.org/fisheries-country-profile-philippines/ Olinayan, A. M. (2014). Conceptual design of a charcoal-fired dryer https://www.researchgate.net/publication/281006323_Conceptual_Design_of_a_Charcoal_fired_dryer Olaniyan, A. M. <i>Department of Agricultural and Bioresources Engineering, Faculty of Engineering, Federal University Oye-Ekiti, P. M. B.303, Oye-Ekiti 371101, Ekiti State, Nigeria; E-mail: amolan397@hotmail.com, adesoji.olaniyan@fuoye.edu.ng</i>

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Patchimaporn Udomkun, Sebastian Romuli, Steffen Schock, Busarakorn Mahayothee, Murat Sartas, Tesfamicheal Wossen, Emmanuel Njukwe, Bernard Vanlauwe, Joachim Müller, Review of solar dryers for agricultural products in Asia and Africa: An innovation landscape approach, Journal of Environmental Management, Volume 268, 2020, 110730, ISSN 0301-4797, <https://doi.org/10.1016/j.jenvman.2020.110730>.

Philippine National Standard PNS/PAES 242:2010 (PAES published 2010) ICS 65.060.01